

张东宽 简历Curriculum Vitae — Dongkuan Zhang

基本信息Personal Information

姓名Name	张东宽Dongkuan Zhang	
国籍Nationality	中国Chinese	
出生日期Date of Birth	02.05.1995	
所在地Location	辽宁大连Dalian, Liaoning, China	
电话Phone	+86 15510689505	
邮箱Email	dkz@mail.dlut.edu.cn	
语言Languages	中文（母语）、英语（流利）Mandarin (Native), English (Fluent)	

教育经历Education

2023 – 今Now	大连理工大学Dalian University of Technology
环境科学与工程，博士（预计 2027 年毕业），导师：姬国钊 教授Ph.D. in Environmental Science and Engineering (expected 2027). Supervisor: Prof. Guozhao Ji	
2020 – 2023	中央民族大学Minzu University of China
环境科学与工程，硕士M.S. in Environmental Science and Engineering	
2013 – 2017	辽宁石油化工大学Liaoning Petrochemical University
石油工程，学士B.S. in Petroleum Engineering	

工作经历Work Experience

2017 – 2019	浙江石油化工有限公司Zhejiang Petrochemical Co., Ltd.
柴油加氢裂化装置操作员。负责装置监控、工艺调整、采样分析、故障处置、安全管理与记录维护。 Diesel hydrocracking unit operator — equipment monitoring, process adjustment, sample collection and analysis, troubleshooting, safety management and record keeping.	

联合培养 / 访学经历Joint Training & Visiting

2025 – 2026	东京科学大学（原东京工业大学）Cross 实验室Cross Laboratory, Institute of Science Tokyo
联合培养博士生，导师：Jeffrey S. Cross 教授。资助：国家公派留学（CSC）。Joint PhD student supervised by Prof. Jeffrey S. Cross. Funded by the China Scholarship Council (CSC).	
2026.09 – 2027.03	卢森堡大学University of Luxembourg
联合培养博士生（6 个月），导师：Bradley P. Ladewig 教授。资助：大连理工大学“连理全球”留学专项。Joint PhD research visit (6 months), supervised by Prof. Bradley P. Ladewig. Funded by the DUT Global Vision Project.	

研究方向Research Interests

- **面向废弃物焚烧技术的智能多相流建模**：将基于物理机理的 CFD 数值模拟与 AI 驱动的流程预测相结合。
Intelligent multiphase flow modeling for waste incineration: combining physics-based CFD with AI-driven flow field prediction.
- 多相流、燃烧与传热过程的欧拉-拉格朗日数值模拟（实验室规模至工业规模）。Eulerian-Lagrangian modeling of multiphase flow, combustion and heat transfer (lab to industrial scale).
- 机器学习与神经网络代理模型（输入凸神经网络等）用于流程快速预测与非牛顿流体建模。Machine-learning and neural-network surrogates (e.g. input-convex neural networks) for fast flow prediction and non-Newtonian fluid modeling.
- 焚烧炉、循环流化床、双流体喷嘴等装备的流程设计、防堵与运行优化。Flow field design, anti-clogging and operational optimization of incinerators, circulating fluidized beds and twin-fluid nozzles.

学术成果概览Academic Output Snapshot

总被引Total citations **60** · h-index **4** · i10-index **1** (更新于updated 2026-09-01)

- 期刊论文Journal articles: **11** 篇papers (其中第一作者 / 共一incl. first/co-first author 5 篇papers)
- 专利Patents: **8** 项patents (含已授权发明专利、实用新型与申请公布granted invention patents, utility models and published applications)
- 软件著作权Software copyrights: **2** 项items
- 团体标准Group standards: **1** 项 (参编) (participating drafter)
- 科技成果转化Technology transfer: **1** 项, 成果转让金额 40 万元patent transfer, RMB 400,000

期刊论文Journal Articles ⁽¹¹⁾

2026	Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow
	D Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G JiD Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G Ji
	<i>Chemical Engineering Research and Design</i> Chemical Engineering Research and Design · link
2026	Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions
	L Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G JiL Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G Ji
	<i>Processes</i> 14 (13), 2189Processes 14 (13), 2189 · link

2026	Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane
	T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ...T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ...
	<i>Industrial & Engineering Chemistry Research</i> <i>Industrial & Engineering Chemistry Research</i> · link
2026	Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies
	T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ...T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ...
	<i>Separation and Purification Technology, 137736</i> <i>Separation and Purification Technology, 137736</i> · link · 2 citations
2025	化工釜残非牛顿流体流变特性的输入凸神经网络建模 Input-convex neural network modeling of rheological properties of non-Newtonian fluids in chemical distillation residues
	张东宽, 褚志强, 田红, 姬国钊Dongkuan Zhang, Zhiqiang Chu, Hong Tian, Guozhao Ji
	<i>西安交通大学学报, 2025, 59(12): 80-91</i> <i>Journal of Xi'an Jiaotong University, 2025, 59(12): 80-91</i> · link
2025	The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator
	J Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G JiJ Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G Ji
	<i>Processes 13 (12), 3928</i> <i>Processes 13 (12), 3928</i> · link
2025	Eulerian-Largrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds Eulerian-Largrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds
	J Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G JiJ Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G Ji
	<i>Applied Thermal Engineering, 127937</i> <i>Applied Thermal Engineering, 127937</i> · link · 5 citations
2025	Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration
	D Zhang, T Anjum, Z Chu, JS Cross, G JiD Zhang, T Anjum, Z Chu, JS Cross, G Ji
	<i>Renewable and Sustainable Energy Reviews 221, 115895</i> <i>Renewable and Sustainable Energy Reviews 221, 115895</i> · link · 39 citations

2023	深海多金属结核采集过程对沉积物扰动试验研究 Experimental study on sediment disturbance during deep-sea polymetallic nodule collection 张东宽, 刘美麟, 夏建新Dongkuan Zhang, Meilin Liu, Jianxin Xia <i>矿冶工程</i> , 2023, 43(3): 20-23 <i>Mining and Metallurgical Engineering</i> , 2023, 43(3): 20-23 · link · 4 citations
2022	深海多金属结核流态化采集头流场参数匹配 Flow field parameter matching of deep-sea polymetallic nodule fluidization acquisition head 张东宽, 官雷, 曹华德, 等Dongkuan Zhang, Lei Guan, Huade Cao, et al. <i>海洋技术学报</i> , 2022, 41(4): 53-60 <i>Journal of Ocean Technology</i> , 2022, 41(4): 53-60 · link · 4 citations
2021	深海多金属结核采集方式及其结构参数比较分析 Comparison and analysis of deep-sea polymetallic nodule collection methods and structural parameters 官雷, 张东宽, 夏玉超, 夏建新Lei Guan, Dongkuan Zhang, Yuchao Xia, Jianxin Xia <i>海洋技术学报</i> , 2021, 40(5): 62-70 <i>Journal of Ocean Technology</i> , 2021, 40(5): 62-70 · link · 3 citations

查看完整出版列表（含引用数） See the full publication list with citation counts

专利Patents ⁽⁸⁾

2026	一种双喷枪自清洁喷雾系统 A dual-spray-gun self-cleaning spray system 姬国钊, 张东宽, 杨君超, 代佳成, 田红, 任桐Guozhao Ji, Dongkuan Zhang, Junchao Yang, Jiacheng Dai, Hong Tian, Tong Ren 中国发明专利申请公布号 CN121972345A (申请号 CN202610181384.4, 2026-05-05 公布) China Invention Patent Application Publication No. CN121972345A (Appl. No. CN202610181384.4, published May 5, 2026)
2026	一种外混式双流体雾化喷枪 An external-mixing dual-fluid atomizing spray gun 姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang 中国发明专利申请公布号 CN121847357A (申请号 CN202610181380.6, 2026-04-14 公布) China Invention Patent Application Publication No. CN121847357A (Appl. No. CN202610181380.6, published Apr 14, 2026)
2025	一种垃圾焚烧炉内自降飞灰的装置 A device for self-settling fly ash in a waste incinerator 姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang 中国实用新型专利, ZL 2024 2 2052228.7 (授权公告号 CN 223137888 U) China Utility Model Patent No. ZL 2024 2 2052228.7 (Granted CN 223137888 U)

2024	一种垃圾焚烧炉内自降飞灰的装置、方法及应用 A device, method and application for self-settling fly ash in a waste incinerator 姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang 中国发明专利申请公布号 CN118912512A (申请号 CN202411163901.2, 2024-11-08 公布) China Invention Patent Application Publication No. CN118912512A (Appl. No. CN202411163901.2, published Nov 8, 2024)
2024	一种基于特斯拉阀的流化床反应器 A fluidized bed reactor based on Tesla Valve 姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang 中国发明专利, ZL202311155946.0 (已授权, 授权公告号 CN117504737B, 2024-10-01) China Invention Patent ZL202311155946.0 (granted; publication No. CN117504737B, Oct 1, 2024) · Scholar
2022	一种多金属结核采集头参数匹配方法 Parameter matching method for polymetallic nodule collection head 夏建新, 官雷, 栾芳芳, 曹华德, 魏定邦, 张东宽Jianxin Xia, Lei Guan, Fangfang Luan, Huade Cao, Dingbang Wei, Dongkuan Zhang 中国发明专利申请公布号 CN114329824A (申请号 CN202111566075.2, 申请公布) China Invention Patent Application Publication No. CN114329824A (Appl. No. CN202111566075.2) · Scholar
2022	一种多金属结核采集头流场设计方法 A design method for flow field of polymetallic nodules collector 夏建新, 张东宽, 栾芳芳, 曹华德, 魏定邦, 官雷Jianxin Xia, Dongkuan Zhang, Fangfang Luan, Huade Cao, Dingbang Wei, Lei Guan 中国发明专利, ZL202111562069.X (已授权) , 申请公布号 CN113946927AChina Invention Patent ZL202111562069.X (granted); application publication No. CN113946927A · Scholar
2016	一种石油勘探用探头 A probe for petroleum exploration 吴成晨, 周博熠, 马硕, 等Chengchen Wu, Boyi Zhou, Shuo Ma, et al. 中国实用新型专利, CN201620273762.3 (已授权, 2016-08-03 授权公告) China Utility Model Patent No. CN201620273762.3 (granted, Aug 3, 2016) · Scholar

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2025-09	双喷枪自清洁控制系统 DSGSCS V1.0 Dual Spray Gun Self-Cleaning Control System (DSGSCS) V1.0 China Computer Software Copyright Registration
2025-07	喷嘴（管道）堵塞风险监测软件 NRMS V1.0 Nozzle (Pipe) Blockage Risk Monitoring Software (NRMS) V1.0 China Computer Software Copyright Registration

标准Standards

2025	《基于高温热解气化的废弃物制氢技术要求》 Technical Requirements for Hydrogen Production from Waste Based on High-Temperature Pyrolysis and Gasification
	张东宽（参编）等Dongkuan Zhang (participating drafter), et al.
	Group/Industry Standard (participating author)

科研项目与科技成果转化Research Projects & Technology Transfer

2023 – 2024	垃圾焚烧炉自降飞灰技术研发（科技成果转化） Self-settling fly ash technology for waste incinerators (technology transfer)
	作为成果完成人之一，将自主研发的专利成果及相关研究成果转让给企业，转让金额 40 万元；个人获学校现金奖励 2 万元。As one of the completing researchers, transferred self-developed patent achievements and related research results to industry (RMB 400,000). Personally awarded RMB 20,000 cash incentive by the university.

荣誉与获奖Honors & Awards

2026.06	首届环境纳米技术国际研讨会（The First International Symposium on Environmental Nanotechnology）最佳墙报奖
	Best Poster Award, The First International Symposium on Environmental Nanotechnology, Dalian, China.
	墙报题目：Poster: <i>Optimizing hazardous waste incineration in a full-scale rotary kiln via co-firing and staged air supply</i> · certificate
2026.04	大连理工大学"连理全球"留学专项资助
	DUT Global Vision Project Funding (University of Luxembourg, 6-month joint PhD visit).
2025.12	教育部 2025 春晖海外青年优秀成果（创新创业类）
	2025 Chunhui Overseas Young Excellent Achievement Collection (MOE of China). Project: "Crisis-Flow Control: Flow-field-based Hazardous Waste Incineration & Disposal Technology and Equipment."
2025.09	博士研究生国家奖学金
	National Scholarship for Doctoral Students (Ministry of Education of China).
2025.11	第 12 届中日韩清洁能源与可持续环境研讨会（日本山梨）报告证书
	12th Japan-China-Korea Joint Symposium on Clean Energy and Sustainable Environment, Yamanakako, Japan — Presentation Certificate.
2025.01	团体标准 T/CITS 242-2025《基于高温热解气化的垃圾制氢技术要求》起草人证书
	Drafter Certificate, Group Standard T/CITS 242-2025, China Inspection and Testing Society.

2024.12	大连理工大学 2024 年度"环境工程之星"
	"Environmental Engineering Star" (Dalian University of Technology).
2024.11	2024 韩中日能源环境联合研讨会 (韩国延世大学) 优秀报告奖
	Award for Excellent Presentation, 2024 Korea-China-Japan Joint Symposium on Energy and Environment (Yonsei University).
2024.11	国家公派留学资格 (CSC 公派证书)
	China Scholarship Council (CSC) Government-Sponsored Studentship (CSC No. 202406060182).
2024	科技成果转化现金奖励 (2 万元)
	Cash incentive for technology transfer (RMB 20,000), Dalian University of Technology.

技术技能Technical Skills

CFD 模拟CFD	ANSYS Fluent、Star-CCM+、COMSOL Multiphysics; 多相流、燃烧、传热建模。ANSYS Fluent, Star-CCM+, COMSOL Multiphysics; multiphase flow, combustion and heat transfer modeling.
编程Programming	Python (含 PyTorch、NumPy、SciPy) 、MATLAB; 数据分析与可视化。Python (PyTorch, NumPy, SciPy), MATLAB; data analysis and visualization.
实验Experiment	流场测试、传热实验、采样与表征。Flow-field testing, heat-transfer experiments, characterization.